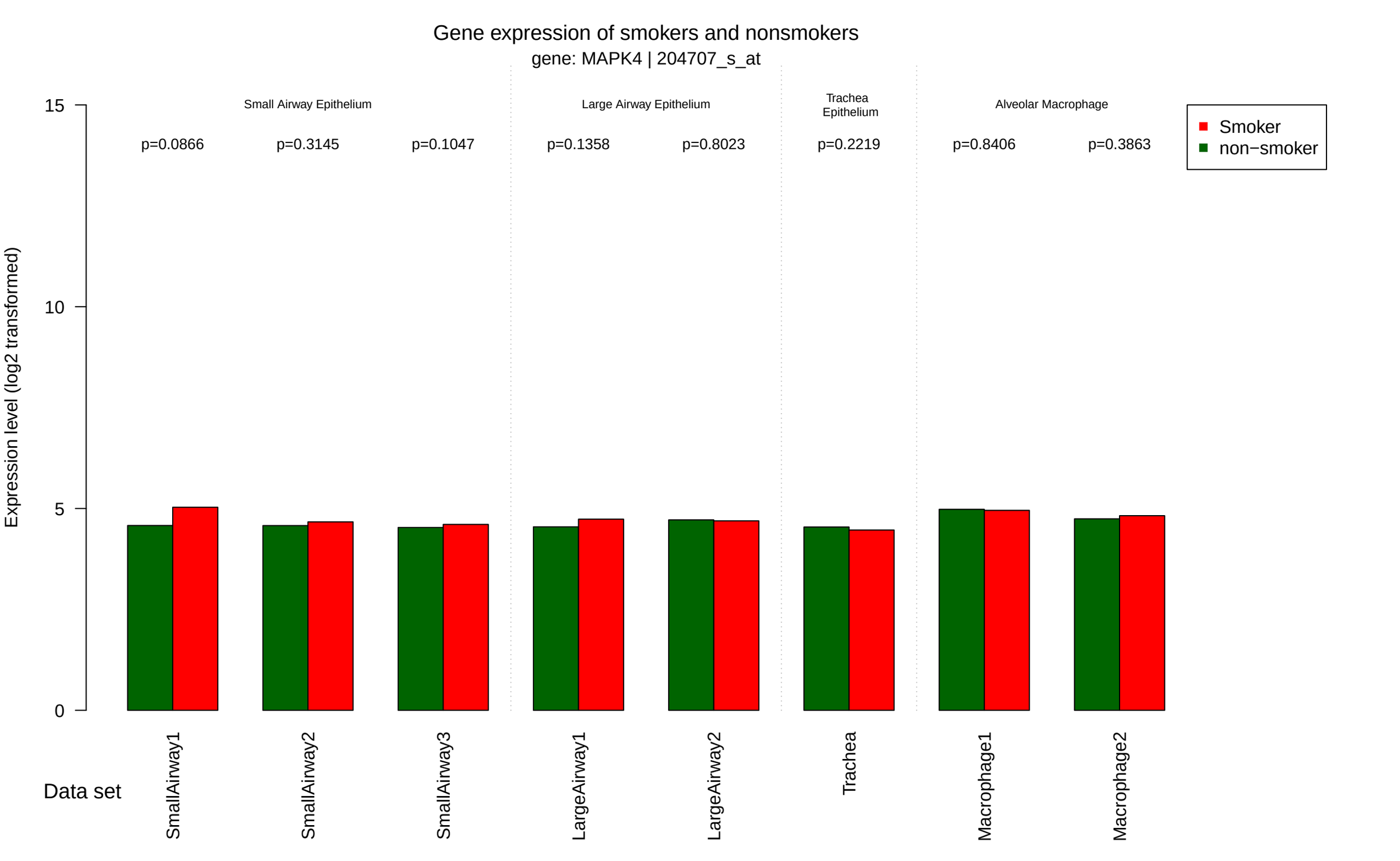
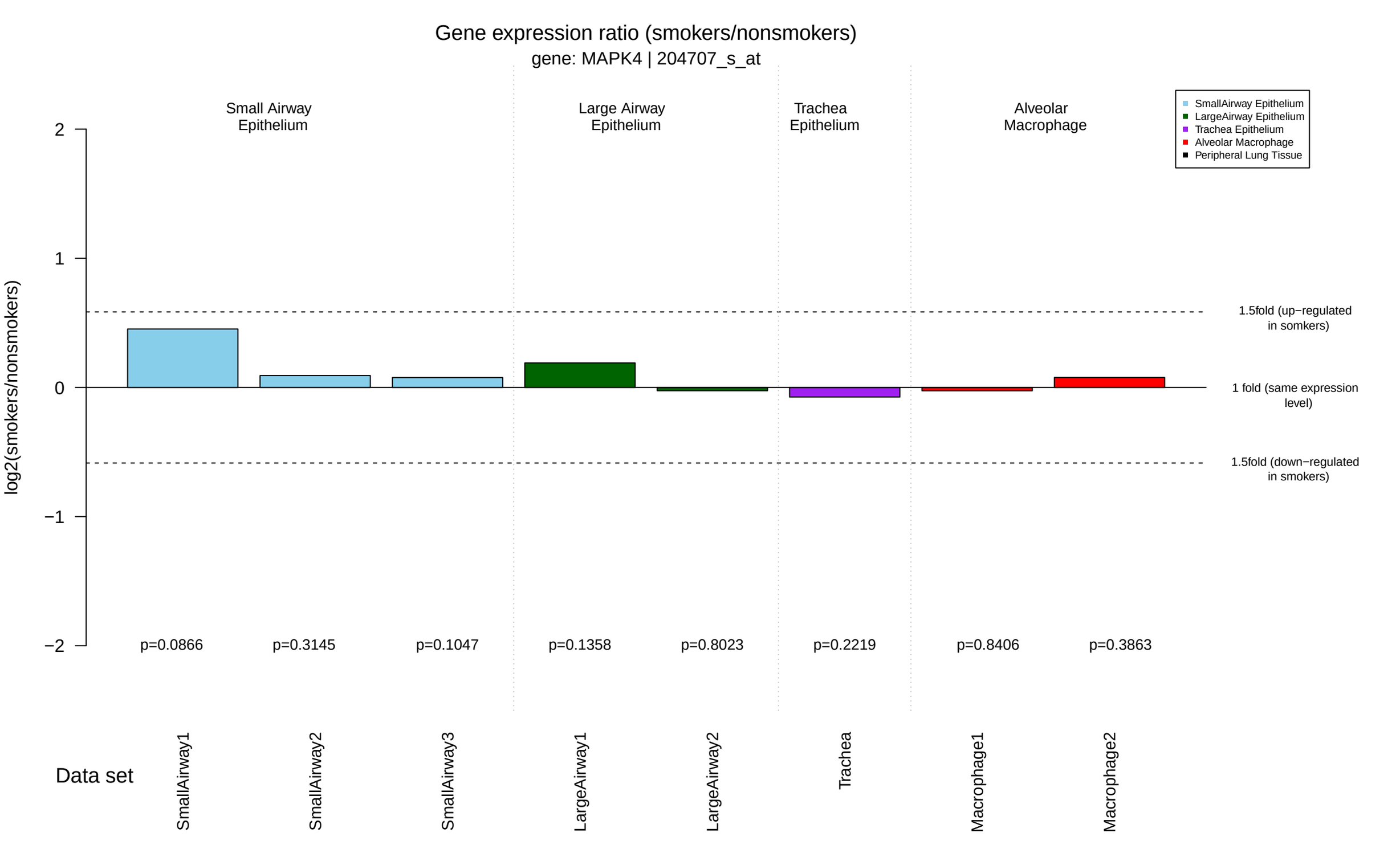
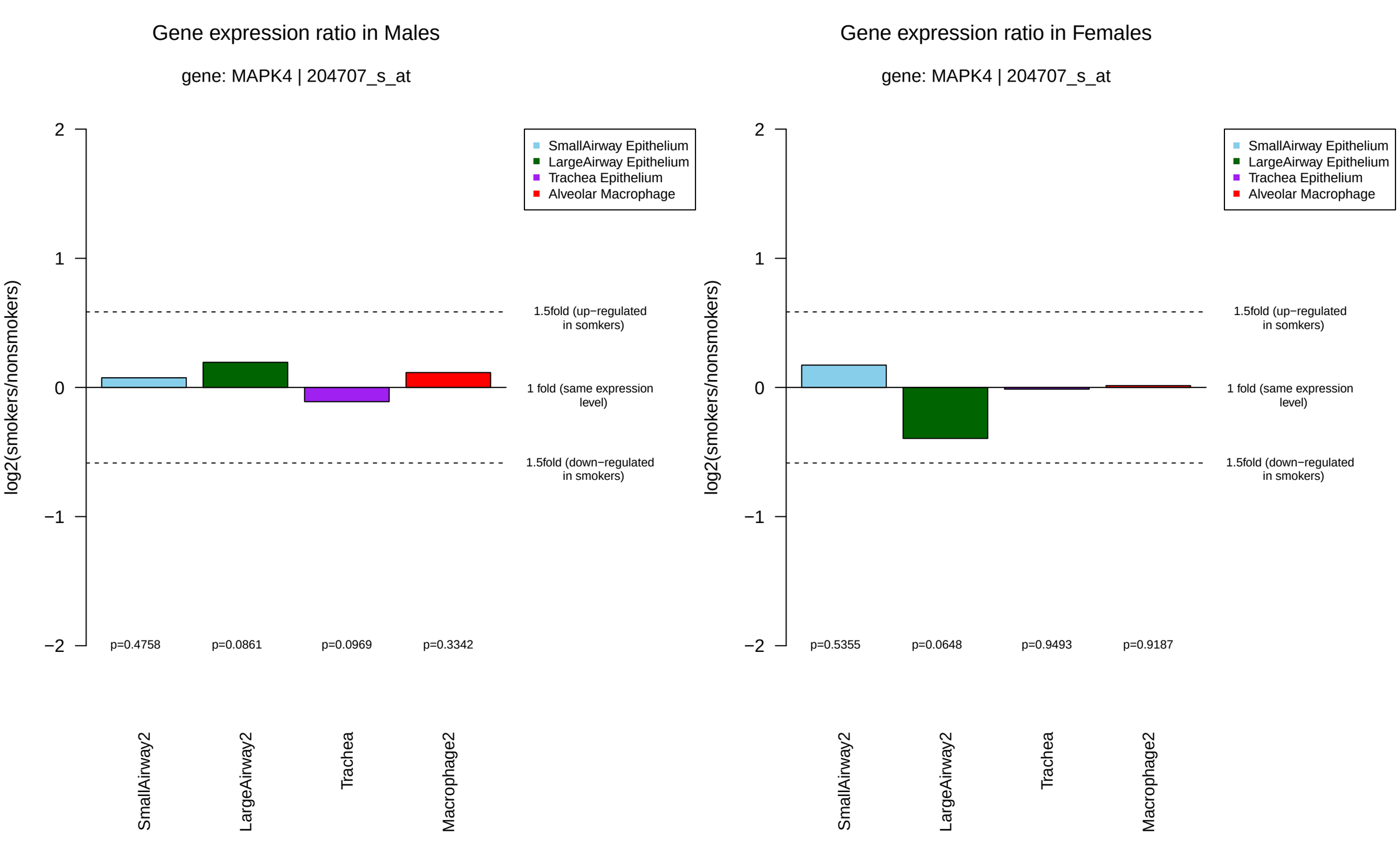
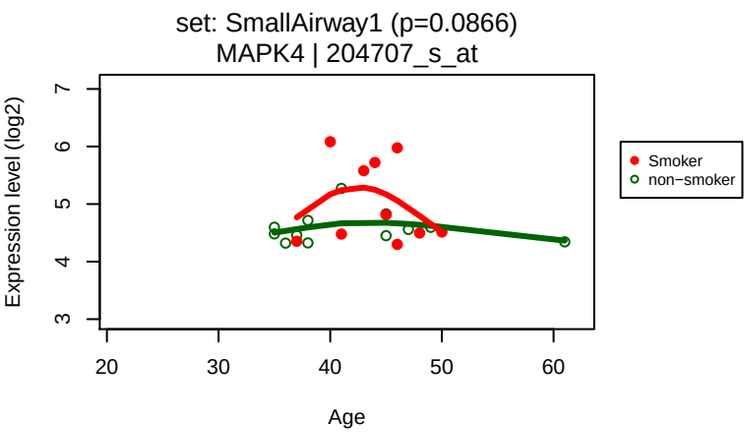
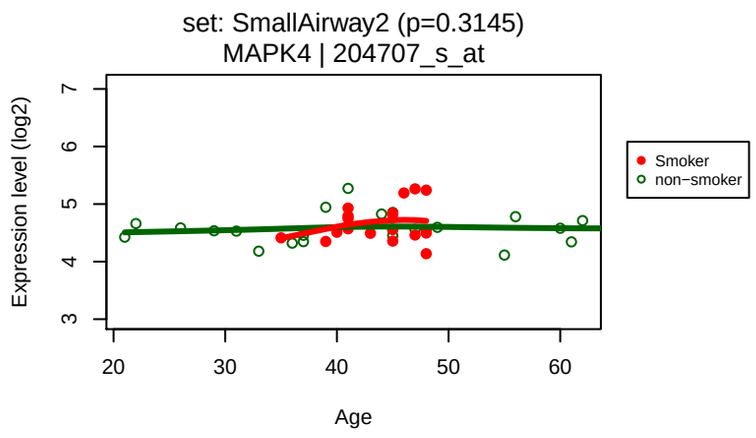
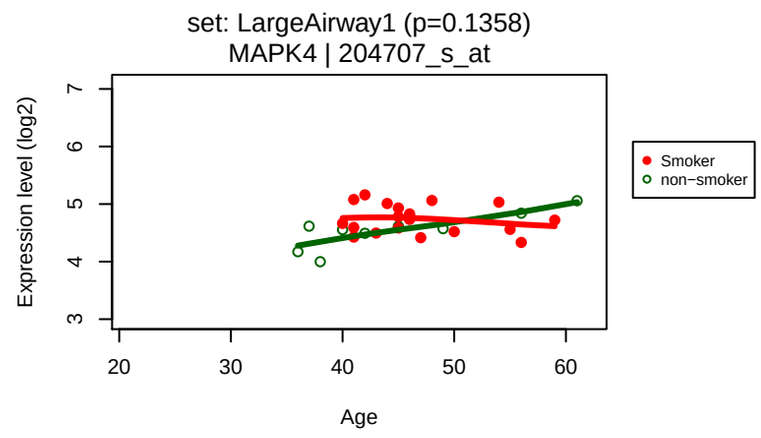

Gene expression of smokers and nonsmokers
gene: MAPK4 | 204707_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0866
p=0.3145
p=0.1047
p=0.1358
p=0.8023
p=0.2219
p=0.8406
p=0.3863
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

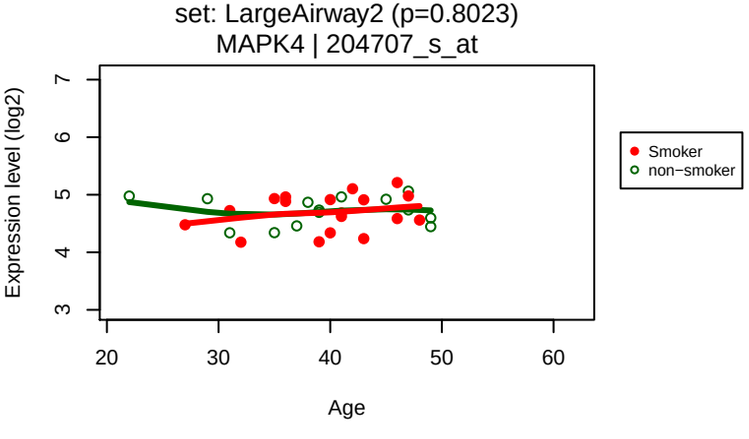

Gene expression ratio (smokers/nonsmokers)
gene: MAPK4 | 204707_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0866
p=0.3145
p=0.1047
p=0.1358
p=0.8023
p=0.2219
p=0.8406
p=0.3863
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

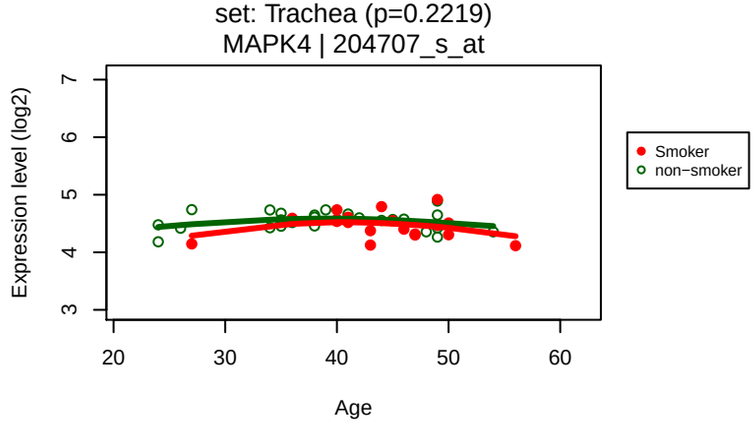

Gene expression ratio in Males
gene: MAPK4 | 204707_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4758
p=0.0861
p=0.0969
p=0.3342
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPK4 | 204707_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5355
p=0.0648
p=0.9493
p=0.9187
SmallAirway2
LargeAirway2
Trachea
Macrophage2

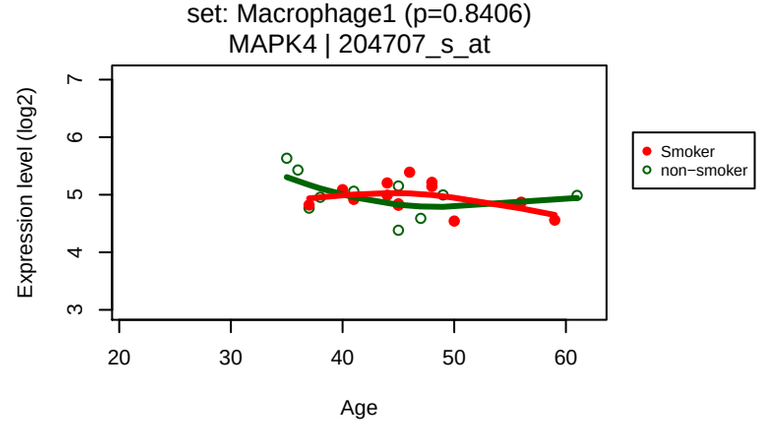

set: SmallAirway1 (p=0.0866)
MAPK4 | 204707_s_at
Expression level (log2)
Age
Smoker
non−smoker

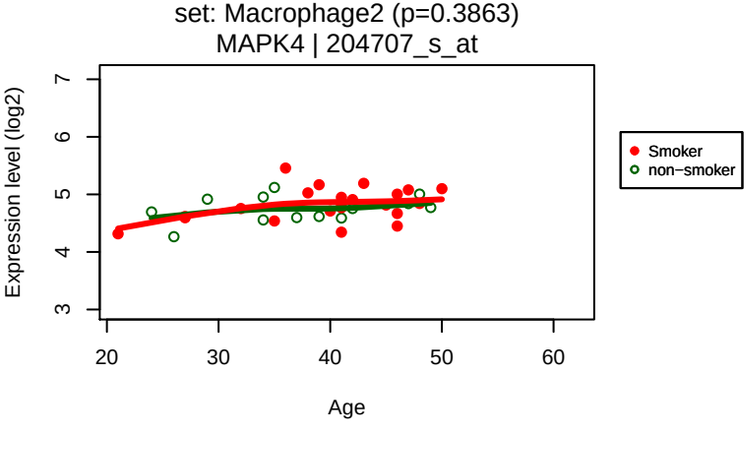

set: SmallAirway2 (p=0.3145)
MAPK4 | 204707_s_at
Expression level (log2)
Age
Smoker
non−smoker

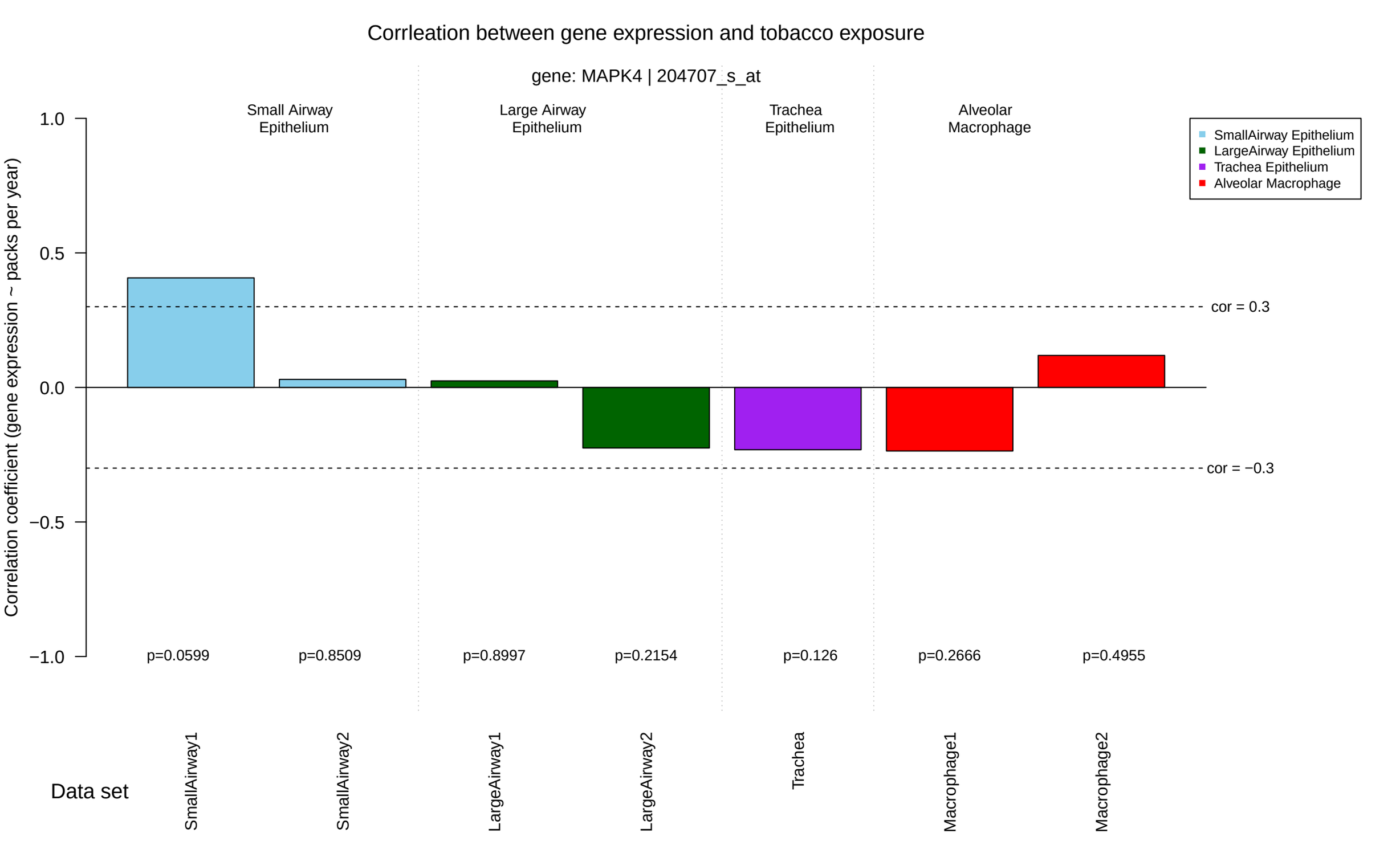

set: LargeAirway1 (p=0.1358)
MAPK4 | 204707_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8023)
MAPK4 | 204707_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2219)
MAPK4 | 204707_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8406)
MAPK4 | 204707_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3863)
MAPK4 | 204707_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPK4 | 204707_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0599
p=0.8509
p=0.8997
p=0.2154
p=0.126
p=0.2666
p=0.4955
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2